ORGANON Crown Width Species Comparison

Considerations for CONUS Applicaion (Working Draft)

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1.0 Introduction

Crown closure is reported often reported by tree-list models. The Forest Vegetation Simulator reports crown closure based on the crown widths of forest grown trees (Crookston and Stage 1999, Wykoff and others 1992). The ORGANON model (https://cips.forestry.oregonstate.edu/organon) compute the crown closure at the tip of the tree (CCH) as a measure of competition above a subject tree [Ritchie and Hann 1990]. The CCH for a tree is computed by taking a horizontal slice of the stand at the top of a subject tree and summing the intersected crown areas of the other trees using crown profile equations.

ORGANON has species specific parameters for maximum crown width (MCW, Paine and Hann 1982) and largest crown width (LCW, Hann 1997) but has 5 sets of parameters for crown profile using height to LCW (HLCW) and crown width above LCW (CWA) (Hann 1999) assigned to species groups Douglas-fir, grand/white fir, western hemlock (incense cedar and red-cedars), pines, and hardwoods for all variants (SWO, NWO, SMC and RAP). The one exception is red alder which uses hardwood parameters for all variants except the RAP variant which has a separate HLCW and CWA models.

In ORGANON CCH is computed using the crown width above (CWA) the height of the large crown width (HLCW). For heights below HLCW, ORANON assigns the LCW for computing crown cover.

Understanding how these crown profile equations compare is important for developing appropriate mappings to other species. Three questions will be investigated in this report:

1. How big are the differences in MCW, LCW, HLCW and CWA equations?
2. How much of the differences in predicted CWA are due to species different MCW, LCW, HLCW and CWA parameters?
3. How much difference do the different stem profile models make in computing CCH?

2.0 Methods

The parameterizations and functions for MCW, LCW, HLCW and CWA from the four ORGANON versions are used (SWO, NWO, SMC, and RAP) that are consistent with version 10 of the ORGANON code. The ORGANON species used here are Douglas-fir (202, DF), grand fir (17, GF), western hemlock (263, WH), ponderosa pine (122, PP), red alder (351, RA) and maple (312, MA). Each of these species has crown profile parameters with maple representing all hardwoods (same HLCW and CWA models), except red alder in the RAP version.

- SWO - Southwestern Oregon

- NWO - Northwestern Oregon
- SMC - Stand Management Cooperative (western OR, WA and BC)
- RAP - Read Alder Cooperative (OR and WA)

For questions 1 and 2 a standard tree will be used to compare differences for the six species.

- DBH = 12.0 inches
- HT = 90.0 feet
- CR = 0.40

Comparisons of crown width will be made at 75-feet above ground.

For question 3 a tree-list will be used for an approximately 22-year old plot from the Panther creek LiDAR study (Flewelling and Marshall 2009). The CCH will be computed for the plot at 25%, 50% and 75% of the height of the tallest tree on the plot by setting all the trees to a single species with crown profile equation and then comparing the differences in CCH for each species crown profile.

3.0 Results

3.1 Compare ORGANON Crown Profile Models

The tables below show the predicted LCW, MCW, HLCW and CWA for DF, GF, WH, PP, RA and MA for each of the four ORGANON versions for the standard tree (see methods). The CWA is at 75 feet.

Table 1: MCW estimates (feet) for ORGANON versions.

SP	MCW.SWO	MCW.NWO	MCW.SMC	MCW.RAP
202	22.54	25.1	25.1	25.1
17	18.27	18.27	18.27	NA
263	21.54	21.54	21.54	21.54
122	18.41	NA	NA	NA
351	26.36	26.36	26.36	31.91
312	31.04	31.04	31.04	31.04

Table 2: LCW estimates (feet) for ORGANON versions.

SP	LCW.SWO	LCW.NWO	LCW.SMC	LCW.RAP
202	18.07	20.2	20.2	20.2
17	16.5	16.5	16.5	NA
263	21.54	17.38	21.54	21.54
122	13.29	NA	NA	NA
351	19.61	19.61	19.61	16.65
312	25.94	25.94	25.94	25.94

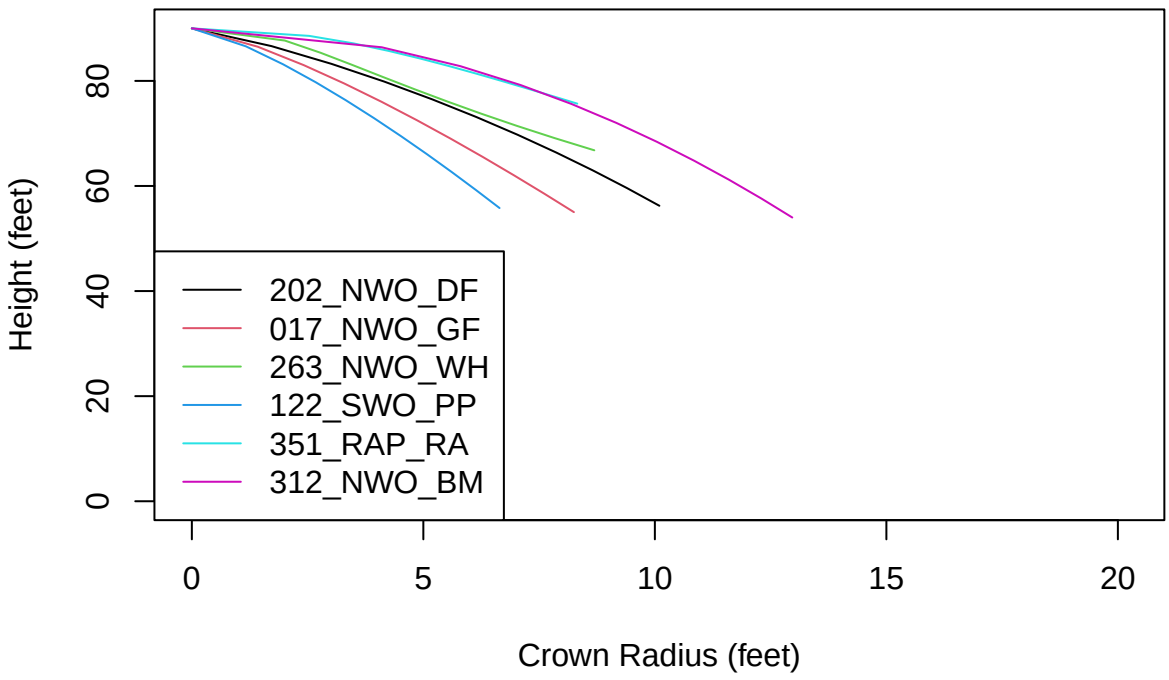
Table 3: HLCW estimates (feet) for ORGANON versions.

SP	HLCW.SWO	HLCW.NWO	HLCW.SMC	HLCW.RAP
202	56.23	56.23	56.23	56.23
17	55.02	55.02	55.02	NA
263	61.55	66.79	61.55	61.55
122	55.8	NA	NA	NA
351	54	54	54	75.67
312	54	54	54	54

Table 4: CWA estimates (feet) for ORGANON versions.

SP	CW75.SWO	CW75.NWO	CW75.SMC	CW75.RAP
202	10.06	11.24	11.24	11.24
17	8.648	8.648	8.648	NA
263	14.4	11.71	14.4	14.4
122	7.131	NA	NA	NA
351	12.66	12.66	12.66	17.14
312	16.74	16.74	16.74	16.74

The graph below shows the crown profiles for DF, GF, WH, PP, RA and MA using the parameters for NWO, except for PP which uses SWO parameters and RA which is from the RAP parameters, for the standard tree (see methods). Taking Douglas-fir as the reference, western hemlock is similar, grand fir and ponderosa pine are narrower and the hardwoods have wider crown widths. Note that the HLCW for the western hemlock and red alder are higher than the other species.



Note that the last table gives an example of differences for the reference tree if we just use DF rather than species group specific parameters.

The tables below look at the differences in the predicted CW at 75-feet (for the standard tree described in the methods) when the MCW and then the LCW and then both MCW and LCW parameters are set to Douglas-fir (using the NWO version). Two tables are shown for each of the three comparisons (MCW, LCW and MCW/LCW). The first graph show the predicted crown widths (feet) and the second shows the difference between using all version and species specific parameters and substituting the Douglas-fir NWO MCW parameter.

This first table shows the differences in crown width at 75-feet for all variants and species and the crown width for Douglas-fir and the NWO variant (Table 4 differences). This is an estimate of the "maximum" expected error of calling everything NWO Douglas-fir (202) and gives a reference to judge how much using variant and species specific MCW and/or LCW reduce these differences.

The comparisons (differences) are give as the default (NWO Douglas-fir CW75) minus the variant and species specific estimate of CW75. Therefor a positive value says using the default gives a larger CW75 estiamte and a negative value says the default estimates a smaller CW75.

Table 5: Difference (feet) of CW estimates using NWO 202 crown profile parameters and versions/species (Table 4) crown profile parametrers (DF - Species_Specific).

SP	DIFF.SWO	DIFF.NWO	DIFF.SMC	DIFF.RAP
202	1.185	0	0	0
17	2.596	2.596	2.596	NA
263	-3.151	-0.4638	-3.151	-3.151

SP	DIFF.SWO	DIFF.NWO	DIFF.SMC	DIFF.RAP
122	4.114	NA	NA	NA
351	-1.415	-1.415	-1.415	-5.898
312	-5.497	-5.497	-5.497	-5.497

Tables comparing the predicted crown width at 75-feet with MCW set to Douglas-fir from the NWO ORGANON version.

Table 6: CW estimates for ORGANON versions with NWO Version MCW parametrers

SP	CW75.SWO	CW75.NWO	CW75.SMC	CW75.RAP
202	11.2	11.24	11.24	11.24
17	11.88	11.88	11.88	NA
263	16.78	13.64	16.78	16.78
122	9.723	NA	NA	NA
351	12.06	12.06	12.06	13.49
312	13.54	13.54	13.54	13.54

Table 7: Difference (feet) of CW estimates using LCW using NWO 202 and versions/species LCW parametrers (DF_LCW - Species_LCW)..

SP	DIFF.NWO	DIFF.RAP	DIFF.SMC	DIFF.SWO
122	NA	NA	NA	2.592
17	3.233	NA	3.233	3.233
202	0	0	0	1.141
263	1.935	2.379	2.379	2.379
312	-3.202	-3.202	-3.202	-3.202
351	-0.6041	-3.655	-0.6041	-0.6041

Tables comparing the predicted crown width at 75-feet with LCW set to Douglas-fir from the NWO ORGANON version.

Table 8: CW estimates for ORGANON versions with NWO Version LCW parametrers.

SP	CW75.SWO	CW75.NWO	CW75.SMC	CW75.RAP
202	10.1	11.24	11.24	11.24
17	7.703	7.703	7.703	NA
263	11.58	11.67	11.58	11.58
122	7.947	NA	NA	NA
351	13.69	13.69	13.69	26.43
312	16.12	16.12	16.12	16.12

Table 9: Difference (feet) of CW estimates using LCW using NWO 202 and versions/species LCW parametrers (DF_LCW - Species_LCW).

SP	DIFF.NWO	DIFF.RAP	DIFF.SMC	DIFF.SWO
122	NA	NA	NA	0.8156
17	-0.9452	NA	-0.9452	-0.9452
202	0	0	0	0.0394
263	-0.0361	-2.814	-2.814	-2.814
312	-0.622	-0.622	-0.622	-0.622
351	1.03	9.287	1.03	1.03

Tables comparing the predicted crown width at 75-feet with MCW and LCW set to Douglas-fir from the NWO ORGANON version.

Table 10: CW estimates for ORGANON versions with NWO Version MCW and LCW parametrers.

SP	CW75.SWO	CW75.NWO	CW75.SMC	CW75.RAP
202	11.24	11.24	11.24	11.24
17	10.58	10.58	10.58	NA
263	13.5	13.6	13.5	13.5
122	10.83	NA	NA	NA
351	13.04	13.04	13.04	20.79
312	13.04	13.04	13.04	13.04

Table 11: Difference (feet) of CW estimates using MCW/LCW using NWO 202 and versions/species MCW/LCW parametrers (DF_MCW/LCW - Species_MCW/LCW).

SP	DIFF.NWO	DIFF.RAP	DIFF.SMC	DIFF.SWO
122	NA	NA	NA	3.704
17	1.935	NA	1.935	1.935
202	0	0	0	1.185
263	1.893	-0.8993	-0.8993	-0.8993
312	-3.705	-3.705	-3.705	-3.705
351	0.3768	3.651	0.3768	0.3768

3.2 Impact of Parameters on CCH calculations

The Panther Creek test plot is about 22 years old at breast height (4.5-feet) and is described in the table below.

Table 12: Panther Creek test plot description: number of trees, minimum DBH (inches), maximum DBH (inches), trees per acre, basal area (ft^2) per acre, quadratic mean DBH (inches), and Lorey mean height (feet).

NTREES	minDBH	maxDBH	TPA	BAPA	QMD	HTlorey
71	28.25	195.3	357.3	190	9.876	72.89

The CCH at 25%, 50% abd 75% of the maximum tree height is computed for the stand as if it were a single species stand of Douglas-fir (NWO), grand fir (NWO), western hemlock (NWO), ponderosa pine (SWO), red alder (RAP) or maple (MA, default hardwood). The results are presented in the table below.

Table 13: Impact in CCH estimates for species using NWO parameters (PP uses SWO and RA uses RAP parameters) for MCW, LCW, HLCW and CWA.

ORGVER	SP	CC25	CC50	CC75
NWO	202	209.8	159.3	11.08
NWO	17	141.8	99.7	7.261
NWO	263	154	154	10.9
SWO	122	91.25	64.02	5.063
RAP	351	155.4	155.4	60.95
NWO	312	324	197.8	17.63

The values generally follow the crown profile graph shown above.

4.0 Discussion

The initial CONUS modeling data set used crown profiles for Douglas-fir for expediency to begin explorations of models since species mappings were not yet developed for all the species in the FIA data set. Based on this analysis it would appear that the major issues is for hardwood stands and stands with hardwoods in the general grown layer which would potentially produce underestimates of CCH.

It is unfortunate there has not been work on crown profiles for more species (especially eastern US) but this is involves challenging data collection. Fortunately, there are more species with MCW and LCW equations to provide estimates of the size of the crown at the it's base and providing some scaling and remove some (not all) of the variation due to mapping (Table 5 vs Table 11).

For mapping species for estimating crown profile, using a conifer and hardwood split would be an improvement over just using the Douglas-fir parameters. Further, it would probably be justified to use separate crown profile equations for pines, firs, Douglas-fir, western hemlock (and "cedars") and hardwoods (RA or hardwood equations). Mapping other species is not as clear. For example, should "unknowns" use default conifer and/or default hardwood parameters?

5.0 References

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